

PUBLICATIONS

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AS FIRST OR CORRESPONDING (*) AUTHOR:

1. **Lu S.***, Frenk C. S., Lacey C. G., et al., *COLIBRE Luminosity Functions – II: the ultraviolet luminosity function from $z = 7$ to $z = 15$ in the COLIBRE simulations*, submitted to MNRAS, [arXiv: 2605.06782](#)
2. **Lu S.***, Frenk C. S., Lacey C. G., et al., *COLIBRE Luminosity Functions – I: from far-UV to submillimetre: luminosity functions at $z = 0$ in the COLIBRE simulations*, submitted to MNRAS, [arXiv: 2605.02022](#)
3. **Lu S.***, Frenk C. S., Bose S., Cole S., Lacey C. G., Baugh C. M., Helly J. C., *A comparison of pre-existing Λ CDM predictions with the abundance of JWST galaxies at high redshift*, 2025, [MNRAS, 536, 1018](#)
4. **Lu S.***, Zhu K., Cappellari M., Li R., Mao S., Xu D., *MaNGA DynPop – V. The dark-matter fraction versus stellar velocity dispersion relation and initial mass function variations: dynamical models and full spectrum fitting of integral-field spectroscopy*, 2024, [MNRAS, 530, 4474](#)
5. **Lu S.***, Zhu K.*, Cappellari M., Li R., Mao S., Xu D., *MaNGA DynPop – II. Global stellar population, gradients, and star-formation histories from integral-field spectroscopy of 10K galaxies: link with galaxy rotation, shape, and total-density gradients*, 2024, [MNRAS, 526, 1022](#)
6. Zhu K.*, **Lu S.***, Cappellari M., Li R.*, Mao S., Gao L., *The MaNGA DynPop – I. Quality-assessed stellar dynamical modelling from integral-field spectroscopy of 10K nearby galaxies: a catalogue of masses, mass-to-light ratios, density profiles and dark matter*, 2023, [MNRAS, 522, 6326](#)
7. **Lu S.***, Xu D.*, Wang S., Wang Y., et al., *Quenched, bulge-dominated, but dynamically cold galaxies in IllustrisTNG and their real-world counterparts*, 2022, [MNRAS, 509, 5062](#)
8. **Lu S.***, Xu D.*, Wang S., Cai Z., et al., *From large-scale environment to CGM angular momentum to star forming activities – II: quenched galaxies*, 2022, [MNRAS, 509, 2707](#)
9. **Lu S.***, Xu D.*, Wang Y., Chen Y., et al., *Hot and counter-rotating star-forming disc galaxies in IllustrisTNG and their real-world counterparts*, 2021, [MNRAS, 503, 726](#)
10. **Lu S.***, Cappellari M., Mao S., Ge J., Li R., *SDSS-IV MaNGA: Stellar population correlates with stellar root-mean-square velocity V_{rms} gradients or total-density-profile slopes at fixed effective velocity dispersion σ_e* , 2020, [MNRAS, 495, 4820](#)
11. **Lu S.***, Xu D., Wang Y., Mao S., et al., *Redshift evolution of the Fundamental Plane relation in the IllustrisTNG simulation*, 2020, [MNRAS, 492, 5930](#)

AS CO-AUTHOR:

1. Gebek A.*, et al. (incl. **Lu S.**), *The COLIBRE-SKIRT pipeline: Calibration-free dust radiative transfer postprocessing for cosmological simulations*, 2026, submitted to A&A
2. Zhong W., Du M.*, **Lu S.**, Jin Y., Zhu K., *Unifying the dynamical classification of early-type galaxies: Kinematic deficits in IllustrisTNG versus observations*, 2026, [A&A, 707, 15](#)
3. Fu Y.*, Cappellari M., Zhu K., Mao S., **Lu S.**, *High-order stellar kinematics in MaNGA integral-field spectroscopy survey: classification, stellar population, and the impact of galaxy bars and mergers*, 2025, [MNRAS, 544, 1038](#)
4. Zhu K.*, Cappellari M., Mao S., **Lu S.** et al., *MaNGA DynPop – VII. A Unified Bulge–Disk–Halo Model for Explaining Diversity in Circular Velocity Curves of 6000 Spiral and Early-type Galaxies*, 2025, [ApJS, 280, 55](#)
5. Wang S.*, Xu D.*, **Lu S.**, *From Larger-scale Cold-gas Angular Momentum Environments to Galaxy Star Formation Activity*, 2025, [ApJ, 986, 85](#)

6. Liu K., Guo H.*, Wang S., Xu D., **Lu S.**, Cui W., Davé R., *Disparate Effects of Circumgalactic Medium Angular Momentum in IllustrisTNG and SIMBA*, 2025, [A&A, 693, 9](#)
7. Wang X.*, Luo Y.*, Faber S. M., et al. (incl. **Lu S.**), *SDSS-IV MaNGA: Stellar rotational support in disk galaxies vs. central surface density and stellar population age*, 2024, [MNRAS, 533, 2026](#)
8. Li S., Li R.*, Zhu K., **Lu S.**, et al., *MaNGA DynPop – VI. Matter density slopes from dynamical models of 6000 galaxies vs cosmological simulations: the interplay between baryonic and dark matter*, 2024, [MNRAS, 529, 4633](#)
9. Chu J.*, Tang H.*, Xu D., **Lu S.**, Long R., *Galaxy stellar and total mass estimation using machine learning*, 2024, [MNRAS, 528, 6354](#)
10. Wang S.*, Xu D., **Lu S.**, Li C., *Kinematical coherence between satellite galaxies and host stellar discs for MaNGA & SAMI galaxies*, 2024, [MNRAS, 527, 7028](#)
11. Wang C.*, Li R.*, Zhu K., et al. (incl. **Lu S.**), *MaNGA DynPop – IV. Stacked total density profile of galaxy groups and clusters from combining dynamical models of integral-field stellar kinematics and galaxy-galaxy lensing*, 2024, [MNRAS, 527, 1580](#)
12. Zhu K.*, **Lu S.**, Cappellari M., Li R.*, Mao S., Gao L., *MaNGA DynPop – III. Accurate stellar dynamics vs. stellar population relations in 6000 early-type and spiral galaxies: fundamental plane, mass-to-light ratios, total density slopes, and dark matter fractions*, 2024, [MNRAS, 527, 706](#)
13. Fu Y.*, Cappellari M., Mao S., **Lu S.**, Zhu K., Li R., *A complete catalogue of broad-line AGNs and double-peaked emission lines from MaNGA integral-field spectroscopy of 10K galaxies: stellar population of AGNs, supermassive black holes, and dual AGNs*, 2023, [MNRAS, 524, 5827](#)
14. Zhu K.*, Li R.*, Cao X., **Lu S.**, Cappellari M., Mao S., *Velocity Dispersion σ_{aper} Aperture Corrections as a Function of Galaxy Properties from Integral-Field Stellar Kinematics of 10,000 MaNGA Galaxies*, 2023, [RAA, 23, 8](#)
15. Abdurro'uf, Accetta K., Aerts C., et al. (incl. **Lu S.**), *The Seventeenth Data Release of the Sloan Digital Sky Surveys: Complete Release of MaNGA, MaStar and APOGEE-2 Data*, 2022, [ApJS, 259, 35](#)
16. Wang S.*, Xu D.*, **Lu S.**, Cai Z., et al., *From large-scale environment to CGM angular momentum to star forming activities – I: star-forming galaxies*, 2022, [MNRAS, 509, 3148](#)
17. Long R. J.*, **Lu S.**, Xu D., *Stellar dynamical modeling - accuracy of 3D density estimation for edge-on axisymmetric galaxies*, 2021, [RAA, 21, 152](#)
18. Xu K., Liu C.*, Jing Y.*, Wang Y., **Lu S.**, *Star Formation in Massive Galaxies at Redshift $z \sim 0.5$* , 2020, [ApJ, 895, 100](#)
19. Li R.*, Li H., Shao S., **Lu S.**, et al., *SDSS-IV MaNGA: the inner density slopes of nearby galaxies*, 2019, [MNRAS, 490, 2124](#)
20. Xu D.*, Zhu L.*, Grand R., Springel V., et al. (incl. **Lu S.**), *A study of stellar orbit fractions: simulated IllustrisTNG galaxies compared to CALIFA observations*, 2019, [MNRAS, 489, 842](#)